

^{133}Xe β^- decay (5.2475 d) [1961Er04](#),[1968A116](#),[1992Ma05](#)

Type	Author	History	Citation	Literature Cutoff Date
Full Evaluation	Yu. Khazov and A. Rodionov, F. G. Kondev		NDS 112, 855 (2011)	31-Oct-2010

Parent: ^{133}Xe : $E=0.0$; $J^\pi=3/2^+$; $T_{1/2}=5.2475$ d 5; $Q(\beta^-)=427.4$ 24; $\% \beta^-$ decay=99.9896 5

[1992Ma05](#): ^{133}Xe gas source; measured E_γ , I_γ , $T_{1/2}$; HPGe detector.

[1961Er04](#): ^{133}Xe source was implanted in Al-backing with isotope separator; measured E_γ , I_γ , E_{ce} , I_{ce} , $\alpha(\text{exp})$, $T_{1/2}$, $Q(\beta^-)$.

Separator, NaI(Tl) detector, double-focusing magnetic spectrometer.

[1968A116](#): ^{133}Xe gas source; measured E_γ , I_γ , $T_{1/2}$; Ge(Li) detector.

Others: [1952Be55](#), [1953Gr07](#), [1958A198](#), [1958PI55](#), [1959Jh17](#), [1960Bo02](#), [1960Mu03](#), [1962Th12](#), [1964Si21](#), [1965Ge04](#), [1969Fr04](#), [1970Av01](#), [1980VyZZ](#).

 ^{133}Cs Levels

E(level) [†]	J^π [‡]	$T_{1/2}$	Comments
0.0	$7/2^+$	stable	
80.9979 8	$5/2^+$	6.283 ns 14	$T_{1/2}$: from Adopted Levels. Measured values in ^{133}Xe β^- decay (5.2475 d) are 6.0 ns 4 (1953Gr07), 6.3 ns 2 (1958A198), 6.25 ns 10 (1962Th12), 6.3 ns 3 (1965Ge04).
160.6121 10	$5/2^+$		
383.8491 8	$3/2^+$		

[†] From a least-squares fit to E_γ 's.

[‡] From Adopted Levels.

 β^- radiations

E(decay)	E(level)	$I\beta^-$ ^{†‡}	Log ft	Comments
(43.6 24)	383.8491	0.0087 10	6.86 9	av $E\beta=11.10$ 63
(266.8 24)	160.6121	1.4 6	7.10 19	av $E\beta=75.16$ 75
346 3	80.9979	98.5 13	5.619 12	av $E\beta=100.62$ 79
E(decay): from 1961Er04 . Other: 347 4 (1952Be55).				

[†] From intensity balances.

[‡] For absolute intensity per 100 decays, multiply by 0.999896 5.

 $\gamma(^{133}\text{Cs})$

I_γ normalization: From $\Sigma(I(\gamma+\text{ce}))=100$ to g.s. and by assuming that there is no direct β^- feeding to the ^{133}Cs ground state.

$I(K\alpha_2 \text{ x ray}/K\alpha_1 \text{ x ray})=0.5310$ 15, $I(K\beta_1 \text{ x ray}/K\alpha_1 \text{ x ray})=0.1767$ 21, $I(K\beta \text{ x ray}/K\alpha \text{ x ray})=0.2021$ 10 ([2007Ya02](#)).

E_γ [‡]	I_γ # [@]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	δ [‡]	α [†]	Comments
79.6142 12	1.2 5	160.6121	$5/2^+$	80.9979	$5/2^+$	M1+E2	0.124 15	1.77 3	$\alpha(K)=1.495$ 22; $\alpha(L)=0.217$ 6; $\alpha(M)=0.0447$ 13; $\alpha(N+..)=0.0107$ 3 $\alpha(N)=0.00940$ 25; $\alpha(O)=0.00128$ 3; $\alpha(P)=5.84 \times 10^{-5}$ 9 $\alpha(L1)=0.1816$ 25; $\alpha(L2)=0.0220$ 21; $\alpha(L3)=0.013$ 3 Mult.: K:L1:L2:L3=0.57 7:0.060 8: ≤ 0.009 : ≤ 0.009 (1969Fr04). δ : Others: <0.16 (1969Fr04); 0.14 3

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$^{133}\text{Xe} \beta^-$ decay (5.2475 d) [1961Er04](#), [1968Al16](#), [1992Ma05](#) (continued) $\gamma(^{133}\text{Cs})$ (continued)

E_γ ‡	I_γ # @	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	δ ‡	α †	Comments
80.9979 11	100	80.9979	5/2 ⁺	0.0	7/2 ⁺	M1+E2	0.158 5	1.703	calculated by evaluators with BrIccMixing program from known shell intensities. I_γ : average of 0.8 1 (1961Er04) and 1.6 7 (1968Al16). $\alpha(\text{K})=1.431$ 20; $\alpha(\text{L})=0.216$ 4; $\alpha(\text{M})=0.0447$ 8; $\alpha(\text{N}+..)=0.01071$ 18 $\alpha(\text{N})=0.00939$ 16; $\alpha(\text{O})=0.001271$ 20; $\alpha(\text{P})=5.56 \times 10^{-5}$ 8 $\alpha(\text{L1})=0.1730$ 24; $\alpha(\text{L2})=0.0253$ 9; $\alpha(\text{L3})=0.0181$ 10 Mult.: K:L1:L2:L3:M1:M2: M3:N1:N2+= 100:12.9 4:1.94 6:1.44 5:3.0 2:0.56 7:0.37 5:0.64 5:0.08 3 averages of 1961Br09 , 1961Er04 , 1964Si21 and 1969Fr04 (evaluated by 1982KhZW). $\alpha(\text{K})=1.43$ 4, weighted average of $\alpha(\text{K})_{\text{exp}}=1.39$ 6 (1961Er04) and 1.46 5 (1970Av01) (absolute K x ray and γ -ray counting). δ : Others: 0.161 9 calculated by evaluators using the BrIccMixing program from shell intensities cited.
160.6120 16	0.2889 23	160.6121	5/2 ⁺	0.0	7/2 ⁺	M1+E2	0.96 5	0.294 6	$\alpha(\text{K})=0.234$ 4; $\alpha(\text{L})=0.0471$ 13; $\alpha(\text{M})=0.0099$ 3; $\alpha(\text{N}+..)=0.00232$ 7 $\alpha(\text{N})=0.00205$ 6; $\alpha(\text{O})=0.000261$ 7; $\alpha(\text{P})=8.08 \times 10^{-6}$ 12 I_γ : from branching ratio in adopted gammas. Others: 0.242 25 (1992Ma05), 0.109 10 (1961Er04) and 0.174 9 (1968Al16). Mult.: ce(K)=0.025 6 (1980VyZZ).
223.2368 13	0.00037 5	383.8491	3/2 ⁺	160.6121	5/2 ⁺	M1+E2	-0.114 14	0.0975	$\alpha(\text{K})=0.0836$ 12; $\alpha(\text{L})=0.01103$ 16; $\alpha(\text{M})=0.00226$ 4; $\alpha(\text{N}+..)=0.000547$ 8 $\alpha(\text{N})=0.000477$ 7; $\alpha(\text{O})=6.64 \times 10^{-5}$ 10; $\alpha(\text{P})=3.26 \times 10^{-6}$ 5 I_γ : from branching ratio in adopted gammas. Others:

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^{133}Xe β^- decay (5.2475 d) [1961Er04](#),[1968Al16](#),[1992Ma05](#) (continued) $\gamma(^{133}\text{Cs})$ (continued)

E_γ [‡]	I_γ ^{#@}	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	δ [‡]	α [†]	Comments
									0.0004 4 (1961Er04), 0.00065 61 (1968Al16) and 0.00044 18 (1992Ma05).
302.8508 5	0.0150 22	383.8491	3/2 ⁺	80.9979	5/2 ⁺	M1+E2	+0.022 20	0.0434	$\alpha(\text{K})=0.0373$ 6; $\alpha(\text{L})=0.00484$ 7; $\alpha(\text{M})=0.000988$ 14; $\alpha(\text{N}+..)=0.000240$ 4 $\alpha(\text{N})=0.000209$ 3; $\alpha(\text{O})=2.92\times 10^{-5}$ 4; $\alpha(\text{P})=1.453\times 10^{-6}$ 21 I_γ : unweighted average of 0.0123 12 (1992Ma05), 0.0135 4 (1961Er04) and 0.0193 7 (1968Al16).
383.8485 12	0.0073 10	383.8491	3/2 ⁺	0.0	7/2 ⁺	E2		0.0202	$\alpha(\text{K})=0.01684$ 24; $\alpha(\text{L})=0.00270$ 4; $\alpha(\text{M})=0.000560$ 8; $\alpha(\text{N}+..)=0.0001326$ 19 $\alpha(\text{N})=0.0001166$ 17; $\alpha(\text{O})=1.541\times 10^{-5}$ 22; $\alpha(\text{P})=5.90\times 10^{-7}$ 9 I_γ : from branching ratio in adopted gammas. Others: 0.00090 4 (1992Ma05), 0.00618 19 (1961Er04) and 0.0062 9 (1961Er04).

[†] [Additional information 1](#).[‡] From adopted gammas. Spectroscopic information on conversion coefficients and subshell ratios from ^{133}Xe β^- decay data is presented. The penetration effect in the conversion was investigated in [1970To01](#), [1975Ra17](#), [1979Th02](#), but the results were contradictory.[#] Normalized to $I_\gamma(80.9974)=100$ by the evaluators.[@] For absolute intensity per 100 decays, multiply by 0.369 3.

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Decay Scheme

Intensities: Relative I_γ

Legend

